EXPERIMENT NO : 9

Name: Sabariprabu .M Roll No : 240701446

Design input forms that validate data (e.g., email, phone number) and display error messages using HTML/CSS, JavaScript

Step 1: HTML Form :

Code :

```
<!DOCTYPE html>
<html>
<head>
   <title>Leave Application Form</title>
</head>
<body>

<h2>Leave Application Form</h2>

<form>
   <label>Name:</label><br>
   <input type="text"><br><br>

   <label>Email:</label><br>
   <input type="text"><br><br>

   <label>Phone Number:</label><br>
```

```
    <input type="text"><br><br>

    <label>From Date:</label><br>
    <input type="date"><br><br>

    <label>To Date:</label><br>
    <input type="date"><br><br>

    <label>Reason:</label><br>
    <textarea></textarea><br><br>

    <button type="submit">Apply Leave</button>
</form>

<p></p>

</body>
</html>
```

Output :

- A simple leave application form is displayed.
- Contains input fields for name, email, phone, dates, and reason.
- A submit button is present.
- No styling and no validation.

Explanation:

- ✓ HTML is used to create the basic structure of the form.
- ✓ <form> collects user input.
- ✓ <input> fields are used for entering data.
- ✓ <textarea> is used for writing the reason.
- ✓ <button> submits the form.
- ✓ At this stage, the form does not check any data.

Output Screen :

Step 2: HTML + CSS :

Code :

```
<!DOCTYPE html>
<html>
<head>
    <title>Leave Application Form</title>

    <style>
        body {
            font-family: Arial;
            background-color: #f2f2f2;
        }

        form {
            background: white;
            padding: 20px;
            width: 350px;
            margin: auto;
        }

        input, textarea {
            width: 100%;
            padding: 8px;
        }

        button {
            background-color: green;
```

```
            color: white;
            padding: 10px;
        }
    </style>
</head>

<body>

<h2>Leave Application Form</h2>

<form>
    <label>Name:</label><br>
    <input type="text"><br><br>

    <label>Email:</label><br>
    <input type="text"><br><br>

    <label>Phone Number:</label><br>
    <input type="text"><br><br>

    <label>From Date:</label><br>
    <input type="date"><br><br>

    <label>To Date:</label><br>
    <input type="date"><br><br>
```

```
    <label>Reason:</label><br>
    <textarea></textarea><br><br>

    <button type="submit">Apply Leave</button>
</form>

<p></p>

</body>
</html>
```

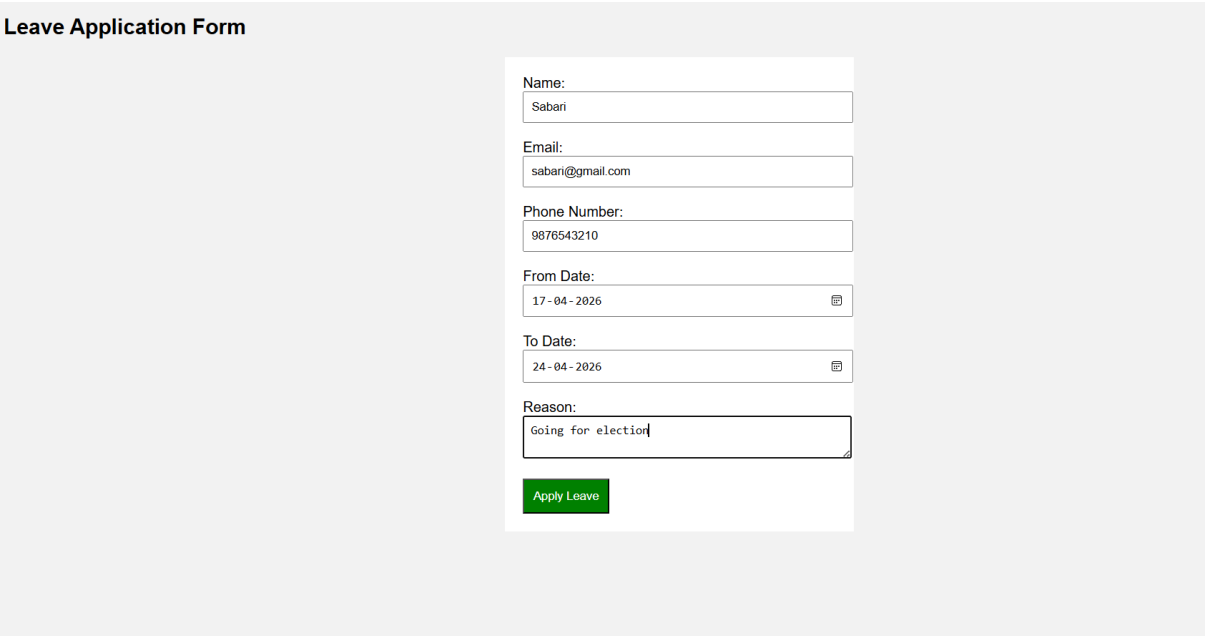
Output :

- The form appears centered on the page.
- Background color changes to light grey.
- Form is displayed inside a white box.
- Button appears in green color.
- Still no validation.

Explanation:

- ✓ CSS is used to style the form.
- ✓ background-color changes the page color.
- ✓ margin: auto centers the form.
- ✓ padding adds spacing inside elements.
- ✓ Improves user interface but does not add functionality.

Output Screen :



Step 3: HTML + CSS + JavaScript :

Code:

```
<!DOCTYPE html>
<html>
<head>
    <title>Leave Application Form</title>

    <style>
        body {
            font-family: Arial;
            background-color: #f2f2f2;
        }

        form {
            background: white;
```

```
            padding: 20px;
            width: 350px;
            margin: auto;
        }

        input, textarea {
            width: 100%;
            padding: 8px;
        }

        button {
            background-color: green;
            color: white;
            padding: 10px;
        }
    </style>
</head>

<body>

<h2>Leave Application Form</h2>

<form id="leaveForm">
    <label>Name:</label><br>
    <input type="text" id="name"><br><br>
```

```
    <label>Email:</label><br>
    <input type="text" id="email"><br><br>

    <label>Phone Number:</label><br>
    <input type="text" id="phone"><br><br>

    <label>From Date:</label><br>
    <input type="date" id="fromDate"><br><br>

    <label>To Date:</label><br>
    <input type="date" id="toDate"><br><br>

    <label>Reason:</label><br>
    <textarea id="reason"></textarea><br><br>

    <button type="submit">Apply Leave</button>
</form>

<p id="error" style="color:red;"></p>

<script
src="https://cdnjs.cloudflare.com/ajax/libs/validator/13.9.0/validator.min.j
s"></script>

<script>
document.getElementById("leaveForm").addEventListener("submit",
function(e) {
```

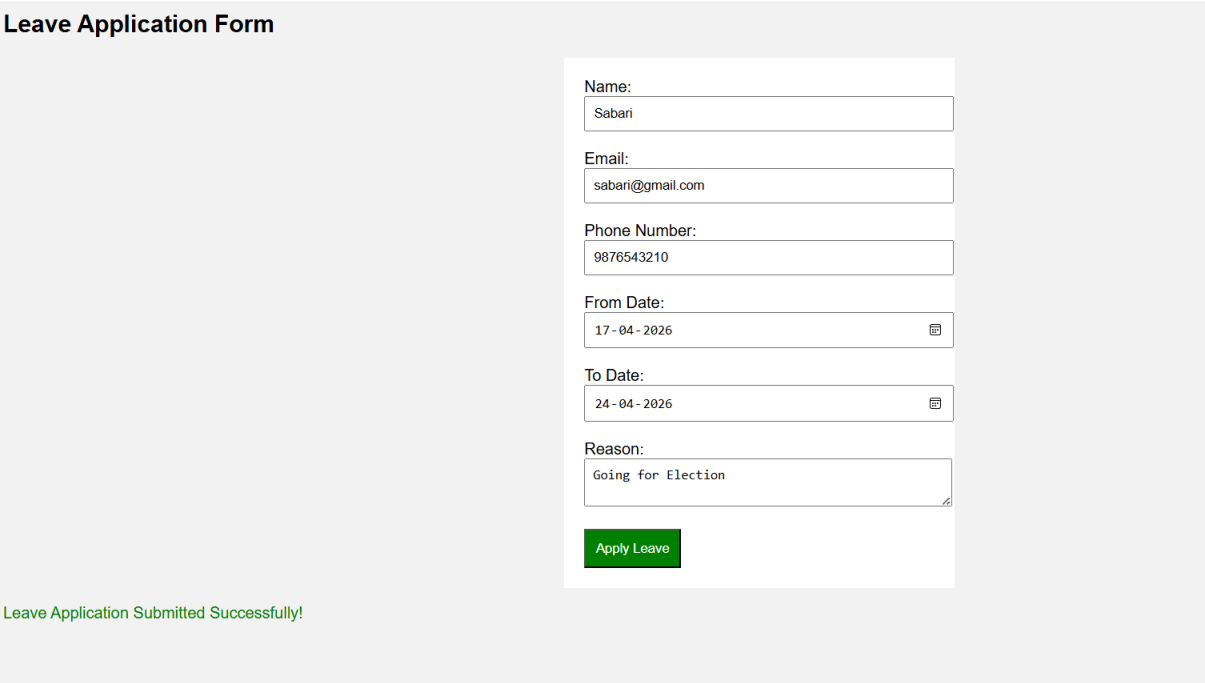
```
e.preventDefault();

let name = document.getElementById("name").value;
let email = document.getElementById("email").value;
let phone = document.getElementById("phone").value;
let fromDate = document.getElementById("fromDate").value;
let toDate = document.getElementById("toDate").value;
let reason = document.getElementById("reason").value;

let error = document.getElementById("error");

error.innerHTML = "";

if (name === "") {
    error.innerHTML = "Name is required!";
    return;
}

if (!validator.isEmail(email)) {
    error.innerHTML = "Invalid Email!";
    return;
}

if (!validator.isMobilePhone(phone, 'any')) {
    error.innerHTML = "Invalid Phone Number!";
    return;
```

```
    }

    if (fromDate === "" || toDate === "") {
        error.innerHTML = "Please select dates!";
        return;
    }

    if (reason === "") {
        error.innerHTML = "Please enter reason!";
        return;
    }

    error.style.color = "green";
    error.innerHTML = "Leave Application Submitted Successfully!";
});
</script>

</body>
</html>
```

Output :

- Form looks same as Step 2.
- When user clicks submit:
  - Shows error messages for invalid inputs.
  - Prevents form submission if data is wrong.
  - Displays success message if all inputs are correct.

Explanation:

- ✓ JavaScript is used for validation.
- ✓ It checks each input field using conditions.
- ✓ Displays error messages using innerHTML.
- ✓ validator.js is used to validate email and phone number.
- ✓ preventDefault() prevents page reload.
- ✓ Adds functionality to the form.

Output Screen: